I. LTI Systems are Characterized by Convolution: Why?

Note: The following is not a rigorous proof.

Recall: (1) We associate δ with a sequence of functions

$$g_1, g_2, \ldots$$

$$g_n(t) = \frac{\sqrt{n}}{\sqrt{2\pi}} \exp\left(-\frac{nt^2}{2}\right)$$

(2) For any sufficiently well-behaved $x(t)$:

$$x = x * \delta = \lim_{n \to \infty} [x * g_n]$$

For any large enough n:

$$x * g_n \approx x$$

Consider any LTI system T:

$$g_n \longrightarrow \boxed{T} \longrightarrow \text{output}$$

Let the output of T as $n \to \infty$ be denoted by h.

$\Rightarrow$ For any large enough n:

$$g_n \longrightarrow \boxed{T} \longrightarrow \approx h$$

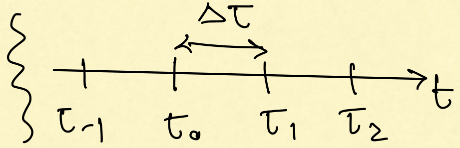
$$\left\{ h = \lim_{n \to \infty} (\text{output}) \right.$$

Now, let us use x as the input to the system T:

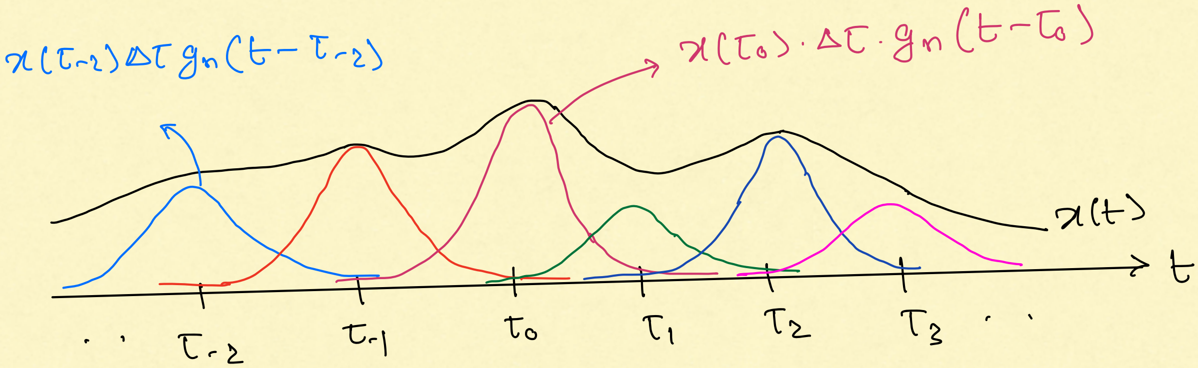
$$x \approx x * g_n$$

$$x(t) \approx \int_{\tau} x(\tau) g_n(t - \tau) d\tau$$

$$\approx \sum_k x(\tau_k)\, g_n(t-\tau_k)\cdot\Delta\tau$$



$$x(t) \approx \sum_k \left(x(\tau_k)\cdot\Delta\tau\right)\cdot g_n(t-\tau_k).$$

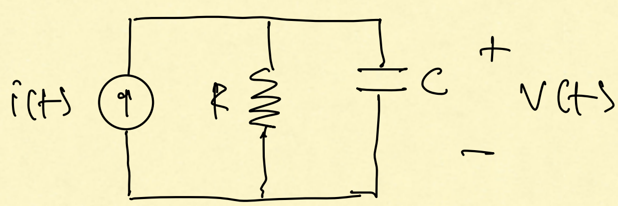


Since T is LTI and $g_n \rightarrow \boxed{T} \rightarrow \approx h$ we deduce that:

$$\sum_k x(\tau_k)\cdot\Delta\tau\cdot g_n(t-\tau_k) \longrightarrow \boxed{T} \longrightarrow \approx \sum_k x(\tau_k)\,\Delta\tau\, h(t-\tau_k)$$

$$\therefore\ y(t) \approx \sum_k x(\tau_k)\cdot h(t-\tau_k)\cdot\Delta\tau \approx \int_{\tau=-\infty}^{+\infty} x(\tau)\, h(t-\tau)\, d\tau = (x*h)(t).$$

II. Impulse Response of an RC Circuit



Solving for $v(t)$ we obtain:

$$v(t) + RC\cdot\frac{dv(t)}{dt} = R\, i(t)$$

$$i(t) \rightarrow \boxed{T} \rightarrow v(t)$$

$$x(t) \rightarrow \boxed{T} \rightarrow y(t).$$

$$y(t) + RC \cdot \frac{dy(t)}{dt} = R \cdot x(t).$$

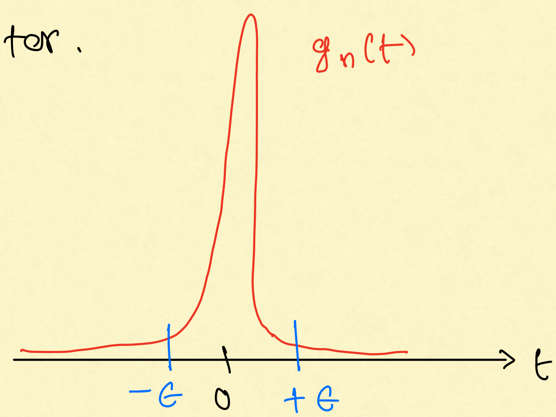
<u>To find impulse response h:</u>

$$g_n \longrightarrow \boxed{T} \longrightarrow \text{output}$$

$$h = \lim_{n \to \infty} (\text{output})$$

<u>Intuition from circuit analysis:</u> (this is <u>NOT</u> a proof).

If n is large, $g_n(t)$ has a narrow and tall peak. Thus, when used as a current waveform, $g_n(t)$ varies rapidly. For a rapidly varying current the capacitor acts like a short circuit, so almost all current flows into the capacitor, with almost nothing flowing through the resistor.



For small $\epsilon > 0$:

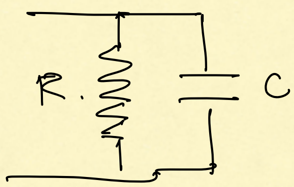
$$\int_{-\epsilon}^{+\epsilon} g_n(t) \approx 1$$

$$\int_{-\epsilon}^{+\epsilon} i(t) \approx 1$$

Almost 1 Coloumb of charge is deposited within a short time across the capacitor.

$\therefore$ At $t = \epsilon$, the capacitor has voltage $\frac{1}{C}$ across its plates.

For $t > \epsilon$, $g_n(t) \approx 0$, i.e. almost no current flows through the current source. The current source acts as a open circuit:



for $t > \epsilon$

$\therefore$ The capacitor slowly discharges through the resistor. This happens exponentially with time constant RC.

Hence, intuitively, we expect that as $n \to \infty$ we must have output:

$$h(t) = \begin{cases} e^{-t/RC} \times \frac{1}{C}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$= \frac{e^{-t/RC}}{C} \cdot u(t).$$